

Topological and Thermodynamic Formalization of Quantitative Proteomics

A Category-Theoretic Approach to Causal Network Inference

Core Philosophy: Objects, Morphisms, and Thermodynamics

Biological systems are not static lists of proteins; they are dynamic, interconnected thermodynamic systems. This pipeline formalizes mass spectrometry data analysis through the lens of structural mathematics. We treat the established evolutionary architecture of the cell (Gene Ontology) as a rigid topological prior—the *spatial manifold*. Experimental quantitative data acts as a *thermodynamic potential* that deforms this space. By breaking the symmetry of the structural prior using the data gradient, we computationally trace the exact directed causal cascade of biological perturbation.

Step 1: The Initial Morphism – Differential Abundance Ranking

To extract stable biological signals from noisy mass spectrometry measurements, the quantified proteome is transformed into a single continuous vector. This morphism condenses the magnitude of the empirical Bayes moderated fold-change and the statistical confidence of the linear model into a unified metric.

$$S_i = \text{sign}(\log_2 \text{FC}_i) \times -\log_{10}(P_i)$$

Taking the negative base-10 logarithm scales high-confidence statistical observations into large magnitudes, sorting the proteome from the most robustly upregulated to the most robustly downregulated driver proteins.

Step 2: Continuous Topology via Gene Set Enrichment Analysis

Rather than relying on arbitrary, discrete significance cutoffs, pathway enrichment topology is evaluated by executing a weighted random walk down the continuous ranked proteome.

$$RES(i) = \sum_{j \leq i, p_j \in K} \frac{|S_j|^p}{N_R} - \sum_{j \leq i, p_j \notin K} \frac{1}{N - |K|}$$

$RES(i)$ is the Running Enrichment Score at rank index i . The maximum absolute deviation from zero isolates the active leading-edge core of proteins driving the biological shift, denoted as $\mathcal{LE}(K)$.

Step 3: Parametric Noise Injection and Stability Index

To combat the heteroskedastic variance inherent in quantitative proteomics, parametric Gaussian noise is injected directly into the effect sizes. Iteratively recalculating the enrichment walk

across these perturbed states quantifies the absolute invariance of the leading-edge proteins, separating true structural drivers from stochastic boundary artifacts.

$$S_{lesi}(p) = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(p \in \mathcal{LE}_b(K)) \quad \text{where} \quad \epsilon_{i,b} \sim \mathcal{N}(0, \eta \cdot SE_i^2)$$

$S_{lesi}(p)$ is bounded between 0 and 1. The perturbation $\epsilon_{i,b}$ is scaled by the protein's empirical standard error SE_i^2 , ensuring that the noise injection respects the instrument's mass measurement uncertainty. The pathway's overall stability is $\bar{S}_{lesi}(K)$.

Step 4: Topological Specificity via Information Content

We quantify the topological specificity of each pathway based on its occurrence probability within the global network, mathematically penalizing redundant, broad functional categories.

$$IC(K) = -\log_2 \left(\frac{|K|}{N_{proteome}} \right)$$

Smaller, highly focused sub-pathways yield large IC values, providing high-resolution functional localization.

Step 5: The Topological Prior – Semantic Similarity Manifold

To compress the highly redundant structure of the Gene Ontology into an orthogonal basis set, we establish the resting state of the biological system. Using the Most Specific Common Ancestor (MICA) between pathway pairs, we calculate the Lin Semantic Similarity matrix $\mathbf{S}_{Lin}$.

$$\mathbf{S}_{Lin}(K_i, K_j) = \frac{2 \cdot IC(\text{MICA}(K_i, K_j))}{IC(K_i) + IC(K_j)}$$

This defines the structural prior—the stage on which the experimental data will act. Clustering operates on the distance matrix $\mathbf{D} = 1 - \mathbf{S}_{Lin}$ via Partitioning Around Medoids (PAM) and Principal Coordinate Analysis (PCoA) to extract independent exemplar medoids representing the $\mathbb{R}^k$ Euclidean manifold.

Step 6: The Thermodynamic Deformation Potential

The experimental data is superimposed onto the topological prior as a scalar field of Potential Energy (U). This metric isolates the absolute biological force exerted on a given node.

$$U(K_i) = |\text{NES}(K_i)| \cdot IC(K_i) \cdot \bar{S}_{lesi}(K_i)$$

Here, $U(K_i)$ acts as the total mass or gravitational pull of pathway K_i . Nodes with massive fold-changes and extreme stability deform the manifold, acting as causal drivers.

Step 7: Symmetry Breaking and the Causal Vector Field

Biological signaling is directional. To infer causal relationships, we break the symmetry of the prior by calculating the thermodynamic gradient between nodes. The final directed edge weight $\mathbf{W}_{i \rightarrow j}$ is the Hadamard product of the data gradient and the structural prior.

$$\Delta U_{ij} = \max(0, U_i - U_j) \quad \Rightarrow \quad \mathbf{W}_{i \rightarrow j} = \Delta U_{ij} \times \mathbf{S}_{Lin}(i, j)$$

This physically restricts the model: energy can only flow downhill ($\Delta U > 0$), and signal can only propagate where an evolutionary biological relationship exists ($\mathbf{S}_{Lin} > 0$). This transforms the undirected hairball network into a Directed Acyclic Graph (DAG).

Step 8: Hyperparameter Optimization via Dirichlet Energy

To mathematically lock the free parameters of the manifold (clusters k_{PAM} and dimensions k_{PCoA}) and prevent subjective overfitting, we calculate the Graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{S}_{Lin}$. The biological realism (smoothness) of the deformation is measured by its Dirichlet Energy across this Laplacian:

$$\mathcal{E}_{Dirichlet} = \mathbf{U}^T \mathbf{L} \mathbf{U} = \frac{1}{2} \sum_{i,j} \mathbf{S}_{Lin}(i,j) (U_i - U_j)^2$$

This equation strictly penalizes steep energy spikes between biologically identical pathways, automatically filtering out technical mass spectrometry noise. The global objective function balances the loss of discrete partition integrity ($\mathcal{L}_{discrete}$ via Silhouette width) against continuous variance and thermodynamic smoothness:

$$\mathcal{L}_{total}(\Theta) = \omega_1(1 - \bar{s}) + \omega_2 \left[\gamma \left(1 - \frac{\sum \lambda_i}{\sum \lambda_{all}} \right) + (1 - \gamma) \left(\frac{\mathbf{U}^T \mathbf{L} \mathbf{U}}{\sum \mathbf{U}^2} \right) \right]$$

Step 9: Topological Inversion and Terminal Phenotype Extraction

To extract the chronological cascade of the organism's response, we must trace the path of *maximum* biological force. Standard graph algorithms (like Dijkstra's) seek the minimum numerical distance. Therefore, an inversion morphism is applied to the edge space to convert massive biological vectors into mathematically "short" distances:

$$\mathcal{W}_{inv} = \max(\mathbf{W}) - \mathbf{W} + \epsilon$$

Applying shortest-path algorithms across $\mathcal{W}_{inv}$ reliably extracts the sequence of events from the highest-energy molecular driver (primary drug target) down to the terminal phenotypic sinks (nodes with an out-degree of zero).